Phase4 Daily Schedule

Phase 4 — Day-by-Day Schedule · Full-Stack / Production AI Systems + MLOps

Weeks 37–42 · Mon May 17 → Sun Jun 27, 2027 · ~165 hrs

Goal: defend every architectural choice and design the next-gen version — LLM internals, senior RAG, MLOps + ML system design, distributed training, serving. Directly reinforces your critic-agent doc-review project.

*Blocks: **A** 06–08 (theory) · **B** 08:30–10:30 (build + threads) · **Evening** 20:30–22:30 (read, Anki, R). Weekend = buffer + rest. Threads: **T** DSA (Block B ~45m) · **M** implement (Block B) · **R** research/apply (Evening).*

Week 37 · May 17–23 — LLM internals (CS336)

Day	Block A	Block B (T/M)	Evening (+ R)
Mon	CS336: tokenization → embeddings	**T:** DP → implement a BPE tokenizer	Anki
Tue	Attention blocks, KV-cache	**M:** implement a KV-cache	Anki
Wed	Sampling: greedy/top-k/nucleus/temp	**T:** heaps → implement samplers	Anki
Thu	Scaling laws; compute budgeting	**M:** a tiny LM training loop	**R:** reproduce-paper
Fri	Efficiency: mixed precision, grad-checkpoint	**T:** timed set	Whiteboard-Fri: token→embedding→blocks→head
Sat–Sun	**Buffer + rest**		

Week 38 · May 24–30 — LLM internals cont + RAG start

Day	Block A	Block B	Evening
Mon	Fine-tuning vs prompting vs RAG	**M:** a minimal retrieval loop	Anki
Tue	Embeddings & vector search	**T:** graphs → build an embedding index	Anki
Wed	Chunking strategies	**M:** chunking + retrieval eval	Anki
Thu	Hybrid search (BM25 + dense)	**T:** strings → hybrid retriever	**R:** write-up
Fri	Reranking	**T:** timed set	Whiteboard-Fri: chunk/embed/retrieve/rerank trade-offs
Sat–Sun	**Buffer + rest**		

Week 39 · May 31–Jun 6 — RAG at senior level

Day	Block A	Block B	Evening
Mon	Hamel/Eugene Yan LLM patterns	**M:** query rewriting + reranker	Anki
Tue	Anthropic contextual retrieval	**T:** DP → contextual chunks	Anki
Wed	Evals: RAGAS, faithfulness	**M:** a RAGAS eval harness	Anki
Thu	DSPy / structured prompting	**T:** graphs → a DSPy pipeline	**R:** reproduce-paper
Fri	Where a critic/verifier improves RAG	**T:** timed set	Whiteboard-Fri: how to evaluate a RAG system honestly
Sat–Sun	**Buffer + rest** · *Deliverable:* RAG system (hybrid + rerank + RAGAS) on public docs		

Week 40 · Jun 7–13 — MLOps + ML systems design

Day	Block A	Block B	Evening
Mon	Chip Huyen: ML systems design framing	**T:** DP → sketch a data/feature layer	Anki
Tue	CS329S: data & feature stores	**M:** a feature pipeline + versioning	Anki
Wed	Training→serving→monitoring loop	**T:** graphs → a serving stub	Anki
Thu	Drift detection; retraining triggers	**M:** a drift-detection check	**R:** write-up
Fri	Made With ML / FSDL patterns	**T:** timed set	Whiteboard-Fri: full train→serve→monitor design

Sat–Sun	**Buffer + rest** · *Deliverable:* an ML-system-design doc (e.g., molecular-property serving)		

Week 41 · Jun 14–20 — Distributed training + inference optimization

Day	Block A	Block B	Evening
Mon	Data vs tensor vs pipeline parallelism	**T:** DP → a DDP toy	Raschka; Anki
Tue	ZeRO / FSDP sharding	**M:** gradient accumulation + AMP	Anki
Wed	vLLM + paged attention (run it)	**T:** graphs → serve with vLLM	Anki
Thu	Quantization (GPTQ/AWQ/GGUF/FP8)	**M:** quantize + benchmark latency	**R:** reproduce-paper
Fri	Throughput vs latency trade-offs	**T:** timed set	Whiteboard-Fri: what ZeRO shards; KV-cache speedups
Sat–Sun	**Buffer + rest**		

Week 42 · Jun 21–27 — Eval / observability / safety + systems-design prep + self-test

Day	Block A	Block B	Evening
Mon	LLM evals; LLM-as-judge pitfalls	**M:** an eval harness (+ judge)	Anki
Tue	Observability (LangSmith/Phoenix)	**T:** graphs → instrument the RAG system	Anki
Wed	Constitutional AI / safety basics	**M:** guardrail checks	Anki
Thu	System-design mock 1 (HCP RAG, 10k/day)	**T:** DP → design artifacts	**R:** write-up
Fri	🏁 **PHASE-4 SELF-TEST** (design an LLM-eval harness; a 45-min ML-system-design whiteboard; defend a serving/monitoring architecture)	**T:** timed set	System-design mock 2 (brain-seg pipeline w/ domain shift)
Sat–Sun	**Buffer + rest**		

End of Phase 4 → Phase 5 (Molecular-ML / Drug-Discovery Track) next — the 15-week differentiator.